

Optimizing Deployment of Irregular Sparse Deep Learning Workloads through ILP-Based Scheduling for Low-Latency Inference

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I. INTRODUCTION

Deep learning models such as convolutional neural networks (CNNs) and Transformers have achieved strong performance in applications including image classification [1], [2], object detection [3], and natural language processing [4]. However, growing model size and complexity continue to increase computational and memory costs, creating challenges for modern accelerators under latency and resource constraints.

A common approach to reducing this cost is sparsity [5]–[7]. In CNNs, pruning removes part of the connectivity between input and output channels. In Transformers, sparse attention restricts token interactions instead of computing all pairwise relationships [8]–[10]. Although these techniques reduce arithmetic work, their practical gains are often limited because sparsity introduces irregular computation patterns that are harder to execute efficiently than dense workloads.

This challenge appears in both CNNs and Vision Transformers [2], [11]. In unstructured group convolutions, irregular channel connectivity can cause load imbalance and repeated data access [6], [12]. In sparse Vision Transformers, attention masks produce blocks that vary in size and structure [9], [10]. Existing high-performance GPU implementations [13]–[15] are mainly designed for dense or fixed tiled execution, and therefore often fail to exploit these irregular sparse patterns effectively. As a result, the benefits of sparsity are often reduced by inefficient scheduling, poor locality, and low hardware utilization [7].

This thesis studies how to execute irregular sparse deep learning workloads more efficiently on modern accelerators. It develops scheduling methods that transform irregular sparse computation into execution plans better aligned with GPU hardware. This idea has been explored in unstructured group convolutions in CNNs and sparse attention in Vision Transformers. In both settings, computation is represented as a matrix structure, reordered to

improve locality, and optimized through profiling-guided scheduling and integer linear programming (ILP). The broader goal is a unified framework for structure-aware and adaptive scheduling of irregular sparse workloads.

II. BACKGROUND AND RELATED WORK

A. Unstructured Group Convolution in CNNs

Convolution is the dominant operation in many convolutional neural networks (CNNs). To reduce cost, many CNN models apply channel pruning [6], [12], which removes some connections between input and output channels. The remaining dependencies can be represented by a connectivity structure indicating which input channels contribute to each output channel. Architectures such as CondenseNet [12] improve efficiency through structured group convolution with contiguous channel groups. After pruning, however, the connectivity pattern may become irregular. These unstructured group convolutions deviate from the regular channel partitions assumed by standard implementations and can reduce efficiency on modern accelerators.

B. Sparse Attention in Vision Transformers

In Transformers, the main computational cost comes from attention, which computes interactions among tokens. Dense attention has quadratic complexity in the number of tokens. To reduce this cost, many Transformer architectures use sparse attention to restrict token interactions. Prior work such as Longformer [9], BigBird [10], and DynamicViT [16] explores sparse patterns based on local attention, global tokens, or token pruning. Although sparse attention reduces arithmetic cost, it often produces irregular computation patterns that are harder to execute efficiently on GPUs.

C. Execution Challenges of Irregular Sparsity

Both unstructured convolution and sparse attention introduce irregular computation structures that can reduce

locality and create workload imbalance. Recent work such as FlashAttention [17] improves attention efficiency through optimized memory access and block-based execution. However, these techniques are mainly designed for dense workloads and may not fully exploit irregular sparse structures. Efficient execution of irregular sparse workloads therefore remains an important challenge in modern deep learning systems.

III. RESEARCH QUESTIONS AND APPROACH

This research aims to improve the execution efficiency of irregular sparse deep learning workloads on modern accelerators, where irregular sparsity often degrades performance through poor load balance, reduced locality, and low hardware utilization. It investigates the following research questions:

- 1) How can irregular sparse computation structures in deep learning models be systematically represented and analyzed?
- 2) How do different sparsity patterns affect workload balance, data locality, and execution efficiency on GPUs?
- 3) How can scheduling decisions for irregular sparse workloads be formulated and solved under hardware resource constraints to improve execution efficiency?

To address these questions, our research proposes sparse computation is represented as a structured matrix, reordered to improve locality and regularity, and optimized through profiling-guided analysis and ILP. Within this framework, ILP models key scheduling decisions, including workload partitioning and execution configuration selection under hardware constraints. Overall, the thesis examines how irregular sparsity shapes GPU execution and develops optimization strategies for more efficient deployment.

IV. CONTRIBUTIONS AND ACHIEVEMENTS

A. Research Results

Currently, this research has produced two main results on the efficient execution of irregular sparse workloads.

The first addresses unstructured group convolutions in CNN inference. By explicitly modeling pruned channel dependencies and optimizing them through partitioning, matrix reordering, and ILP, the proposed method achieved up to $1.9\times$ speedup over PyTorch Conv2D on ShuffleNet and CondenseNet, with the best variant reaching $2.0\times$ on CondenseNet-4 and $1.75\times$ on ShuffleNet while preserving Top-1 accuracy.

The second addresses sparse attention in Vision Transformers. By reorganizing sparse attention structures and applying offline profiling with ILP-based scheduling, the framework achieved up to $2.13\times$ end-to-end speedup over fixed-tile FlashAttention. Under larger resolutions

and smaller patch sizes, the speedup further increased to $4.6\times$ for Vision Longformer, $4.4\times$ for RegionViT, and $4.2\times$ for DynamicViT, without accuracy loss.

These results show that structure-aware modeling and optimization-based scheduling can substantially improve the execution efficiency of irregular sparse workloads across both CNN and Transformer architectures.

B. Dissemination and Community Engagement

The research outcomes have been disseminated through peer-reviewed publications and conference presentations.

- **HiPC 2025** – Paper: *Optimizing Deployment of Unstructured Group Convolutions for Low Latency Inference*
- **IPDPS 2026** – Paper: *Achieving Low Latency Inference on High Resolution Images by Exploiting Sparsity in Vision Transformers*

V. SUMMARY AND OUTLOOK

Current results show that structure-aware scheduling improves the execution efficiency of irregular sparse workloads in both CNNs and Vision Transformers. By representing sparse computation structures and applying reordering with ILP-based scheduling, the framework improves load balance and hardware utilization. However, several challenges remain before a general scheduling framework for sparse deep learning workloads can be established.

First, the current scheduling strategies rely mainly on offline profiling. Although this approach identifies efficient execution configurations, it does not fully adapt to runtime variation in sparsity patterns and workload characteristics. Future work will investigate adaptive scheduling strategies that select execution configurations dynamically from runtime information, including lightweight cost models for estimating the efficiency of different execution strategies under GPU resource constraints. Second, the current framework focuses on two representative model classes: CNNs and Vision Transformers. Although these cover many sparse computation patterns, future work will explore how the framework can be generalized to additional sparse workloads and whether unified scheduling abstractions can be developed for different sparse computation structures. Finally, we will study the scalability of the framework on larger models and more diverse hardware platforms, including large GPU clusters and emerging accelerator architectures.

The long-term goal is to develop a general scheduling framework for irregular sparse deep learning workloads that systematically improves execution efficiency across model architectures and hardware platforms.

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